$$c=(d,b,e,t),\qquad 0\le b<e\le|x|. \tag{2-1}$$

$$\rho_L=b-a_s,\quad \rho_R=z_u-e;\qquad b=a_s+\rho_L,\quad e=z_u-\rho_R. \tag{2-2}$$

$$\text{Attention}(Q,K,V)=\text{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V. \tag{2-3}$$

$$P_\theta(Y|x)=\prod_{k=1}^{L}P_\theta(y_k|x,y_{<k}), \tag{2-4}$$

$$L_{\text{gen}}=-\sum_{k=1}^{L}\log P_\theta(y_k|x,y_{<k}). \tag{2-5}$$

$$W=W_0+\frac{\alpha}{r}BA,\qquad A\in\mathbb{R}^{r\times d_{\text{in}}},\quad B\in\mathbb{R}^{d_{\text{out}}\times r}. \tag{2-6}$$

$$\text{JS}(p,q)=\frac{1}{2}\text{KL}(p\|m)+\frac{1}{2}\text{KL}(q\|m),\qquad m=\frac{p+q}{2}. \tag{2-7}$$

$$G_{\text{rank}}=\text{argmax}_{G\in S}E_{\text{rank}}(G),\qquad G_0=L_A(G_{\text{rank}}). \tag{2-8}$$

$$\text{Precision}=\frac{TP}{TP+FP},\quad \text{Recall}=\frac{TP}{TP+FN},\quad F_1=\frac{2TP}{2TP+FP+FN}. \tag{2-9}$$

$$p=\frac{1+\sum_{b=1}^{B}1(|\Delta_b^*-\hat{\Delta}|\ge|\hat{\Delta}|)}{B+1}. \tag{2-10}$$

$$h'_t=W_0h_t+\frac{\alpha}{r}B_gA_gh_t+g_t\frac{\alpha}{r}B_cA_ch_t,A_\cdot\in\mathbb{R}^{r\times d_{in}},B_\cdot\in\mathbb{R}^{d_{out}\times r}. \tag{3-1}$$

$$L_{gen}(x,G)=-\frac{1}{T_y}\sum_{t=1}^{T_y}\log p_\theta(y_t|x,y_{<t}). \tag{3-2}$$

$$L_{sent}=-\frac{1}{N_s}\sum_{i=1}^{N_s}\left[z_i\log\hat{z}_i+(1-z_i)\log(1-\hat{z}_i)\right],L_{ICIC}=L_{gen}+\lambda_{sent}L_{sent}. \tag{3-3}$$

$$D_A=\{\omega\in\Omega_{sem}(C):V(\omega)\cap A\neq\varnothing\}. \tag{3-4}$$

$$\varphi:C\to C',\quad (d,b_c,e_c,t_c)\mapsto(d',b_c',e_c',t_c). \tag{3-5}$$

$$R_A\subseteq J_A\subseteq D_A\subseteq\Omega_{sem},\quad \Omega_{sem}=\Omega\mid U_{sem}. \tag{3-6}$$

$$u_c^x=MeanPool_{i\in B_x(c)}h_i^x,\ p_C^x(c)=softmax\,f_C\left(\left[u_c^x;h_{end}^x\right)\right). \tag{3-7}$$

$$p_R^x(S,v)=softmax\,f_R\left(\left[Pool\left(\{u_c^x:c\in S\}\right);u_v^x;h_{end}^x;\eta(S,v)\right)\right), \tag{3-8}$$

$$A^+=A\cup\bigcup_{\substack{\omega\in D_A\\ y_\omega\neq NONE}}V(\omega). \tag{3-9}$$

$$L_C^{align}(x,x')=\frac{1}{|A^+|}\sum_{c\in A^+}JS\left(p_C^x(c),p_C^{x'}(\phi(c))\right), \tag{3-10}$$

$$L_R^{align}(x,x')=\frac{1}{|B|}\sum_{b\in B}\frac{1}{|D_b|}\sum_{\omega\in D_b} JS\left(p_R^{x}(\omega),p_R^{x'}(\phi(\omega))\right). \tag{3-11}$$

$$S(x;M_C,M_R)=\frac{1}{|M_C|}\sum_{c\in M_C} CE\left(p_C^{x}(c),t_c\right)+\frac{1}{|B_M|}\sum_{b\in B_M}\frac{1}{|M_R^b|}\sum_{\omega\in M_R^b} CE\left(p_R^{x}(\omega),y_\omega\right), \tag{3-12}$$

$$L_{heads,d}^{sup}=\frac{S(x_d,G_d)+\sum_{j=1}^{m_d} S_{m_j}\left(x'_{dj},\phi_j(G_d)\right)}{1+m_d}. \tag{3-13}$$

$$L_d=L_{ICIC}(x_d,G_d)+\lambda_{sup}L_{heads,d}^{sup}+\lambda_C\acute{L}_{C,d}^{align}+\lambda_R\acute{L}_{R,d}^{align}. \tag{3-14}$$

$$C=\{C^{(1)},\ldots,C^{(K)}\}. \tag{4-1}$$

$$E_{rank}(G)=0.4\,E_{dens}(G)+0.4\,E_{coh}(G)+0.2\,E_{val}(G). \tag{4-2}$$

$$E_{dens}(G)=\frac{1}{|Q(G)|}\sum_{j\in Q(G)} min\left\{\frac{d_{coh}^{in}(j)}{\tau_j},1\right\},E_{coh}(G)=\frac{1}{|R|}\sum_{r\in R} p_r. \tag{4-3}$$

$$q_C(c\mid x),q_R(r\mid x). \tag{4-4}$$

$$\Phi_X(G)=2\sum_{o\in X(G)} q_X(o\mid x)-\kappa_X|X(G)|,\Phi(G)=w_C\Phi_C(G)+w_R\Phi_R(G), \tag{4-5}$$

$$\Phi_X(G)=(2-\kappa_X)E\left[|Y_X|\mid x\right)-E\left[\kappa_X F P_X+(2-\kappa_X)F N_X\mid x\right). \tag{4-6}$$

$$\Delta\Phi_C\geq 0,\Delta\Phi_R\geq 0,\Delta\Phi>\delta>0. \tag{4-7}$$

$$D=D_0\cup\{c\in C:\exists b\in B,Conflict(c,b)\},C'=(C\setminus D)\cup B. \tag{4-8}$$

$$P_{tx}=P_A\cup S_R. \tag{4-9}$$

$$\max_y w_C\Phi_C(C')+w_R\sum_{r\in P}(2q_R(r\mid x)-\kappa_R)y_r. \tag{4-10}$$

$$\begin{gathered}\sum_{t\in T(u)} z_{(u,t)}\leq 1,\\ z_c+z_d\leq 1\text{若}Conflict(c,d);\\ y_r\leq z_c\,\forall c\in Ends(r),\\ \sum_{r\in\Gamma_s} y_r\leq 1\,\forall s;\\ y_r=I[r\in R)\,\forall r\notin P_{tx},\\ G'\in G_A.\end{gathered} \tag{4-11}$$

$$o_v\geq o_u+1-M\left(1-y_{(u,v,\ell)}\right). \tag{4-12}$$

$$d_R=\sum_{r:y_r^0=1}(1-y_r)+\sum_{r:y_r^0=0} y_r. \tag{4-13}$$

$$\max_{z,y}\Phi(z,y)=w_C\sum_{v\in V} a_C(v)z_v+w_R\sum_{r\in P} a_R(r)y_r. \tag{4-14}$$

$$\sum_{t\in T_*(u)} z_{(u,t)} \le 1 \,\forall u \in U,$$
$$z_c + z_d \le 1 \text{若} \, Conflict(c,d);$$
$$y_r \le z_c \,\forall r \in P, \forall c \in Ends(r), \tag{4-15}$$
$$\sum_{r\in\Gamma_s} y_r \le 1 \,\forall s;$$
$$y_h + z_w \le 1 \,\forall h \in P_{CDCP}, \forall w \in Btw(h).$$

$$d_C^{net} = \sum_{v:z_v^0=1} (1-z_v) + \sum_{v:z_v^0=0} z_v \le B_C, d_R^{net} = \sum_{r:y_r^0=1} (1-y_r) + \sum_{r:y_r^0=0} y_r \le B_R. \tag{4-16}$$

$$\Delta\Phi_X = \sum_{o\in X(G')\setminus X(G)} a_X(o) - \sum_{o\in X(G)\setminus X(G')} a_X(o). \tag{4-17}$$

$$F' - F = \frac{2\Delta T - F\Delta P_0}{P_0 + Q + \Delta P_0}. \tag{4-18}$$